

Bryan Torres

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech)

Master of Engineering in Computer Science & Applications

Blacksburg, Virginia

Expected May 2027

GPA: 3.9 / 4.0 | **Concentrations:** Computer Systems & Architecture; Software Systems

Virginia Polytechnic Institute and State University (Virginia Tech)

Bachelor of Science in Computer Science

Blacksburg, Virginia

Aug. 2022 – May 2026

EXPERIENCE & RESEARCH

Software Engineer Intern | Expedition Technology

Geospatial Dataset Diversity

May 2026 – Aug. 2026

Herndon, Virginia

- Engineered a 0–1 diversity-scoring framework for large-scale geospatial imagery datasets, defining and feature-engineering four dimensions: geographic, temporal, illumination, and image quality.
- Applied normalized Shannon entropy to quantify distribution evenness across geographic and temporal coverage, including country, region, biome, settlement, time-of-day, month, season, and interval features.
- Implemented Python/FastAPI backend scoring and CRUD workflows and integrated them with an existing TypeScript/Vite frontend in a production monorepo.
- Built interactive Chart.js and TopoJSON/GeoJSON visualizations, including a geographic heat map, bar and doughnut charts, and a radar chart for feature-level diversity scoring and recommendations.
- Integrated work into company delivery workflows by adapting reusable CI templates and creating a Helm/Argo CD pre-deployment database migration job with dependency-aware sync ordering.

Software & DevOps Engineer Intern | Dark Wolf Solutions

Project Syndicate

Jun. 2025 – Aug. 2025

Herndon, Virginia

- Built and deployed “Syndicate,” a cloud-native CI/CD framework that converts customer CI YAML into JSON for automated vulnerability scanning across pipelines.
- Authored Kubernetes manifests and Argo CD applications; deployed a Next.js frontend and FastAPI services to Google Kubernetes Engine.
- Implemented Dex/OIDC authentication and integrated SonarQube and OWASP dependency checks into GitLab CI for automated pre-merge security and quality gates.
- Wrote Jest and PyTest/Pylint test suites, raising overall test coverage to 80%; containerized services with Docker and reduced image size by 22%.

Undergraduate Researcher | BURGs, Virginia Tech | MARCO

Machine Learning Performance Group

Oct. 2024 – Dec. 2025

Blacksburg, Virginia

- Conducted research on LLM-driven code generation and optimization for high-performance matrix multiplication.
- Developed a LangChain-based multi-agent system with retrieval-augmented generation and prompt strategies for higher-quality optimized code.
- Led benchmarking of LLM API services and selected o3-mini based on token efficiency and task accuracy, reducing experiment cost by 78%.
- Co-author: “MARCO: Multi-Agent Code Optimization with Real-Time Knowledge Integration for High-Performance Computing,” arXiv:2505.03906 (2025).

TECHNICAL SKILLS

- Languages:** Python, TypeScript, JavaScript, C, Java, C#
- Frameworks & Data:** FastAPI, React, Vite, Next.js, Nodfe.js, Chart.js, SQLAlchemy, Alembic, PostgreSQL, GeoJSON/TopoJSON
- DevOps & Cloud:** Docker, Kubernetes, Helm, Argo CD, GitLab CI/CD, Google Cloud Platform, SonarQube, Dex/OIDC
- AI/ML & Tools:** PyTorch, Hugging Face, LangChain, OpenAI/Anthropic APIs, Ollama, Git/GitHub, Linux/Unix, tmux